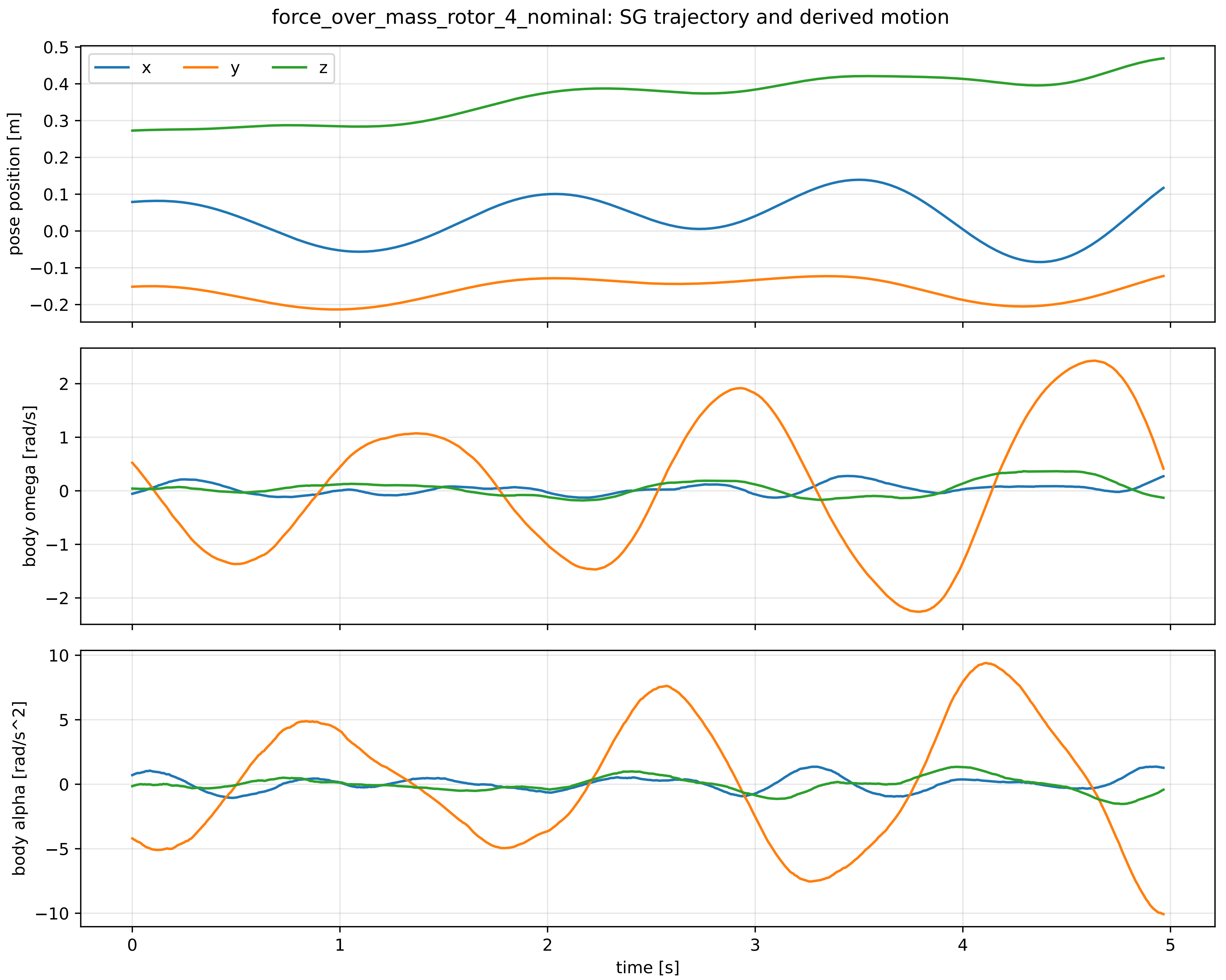
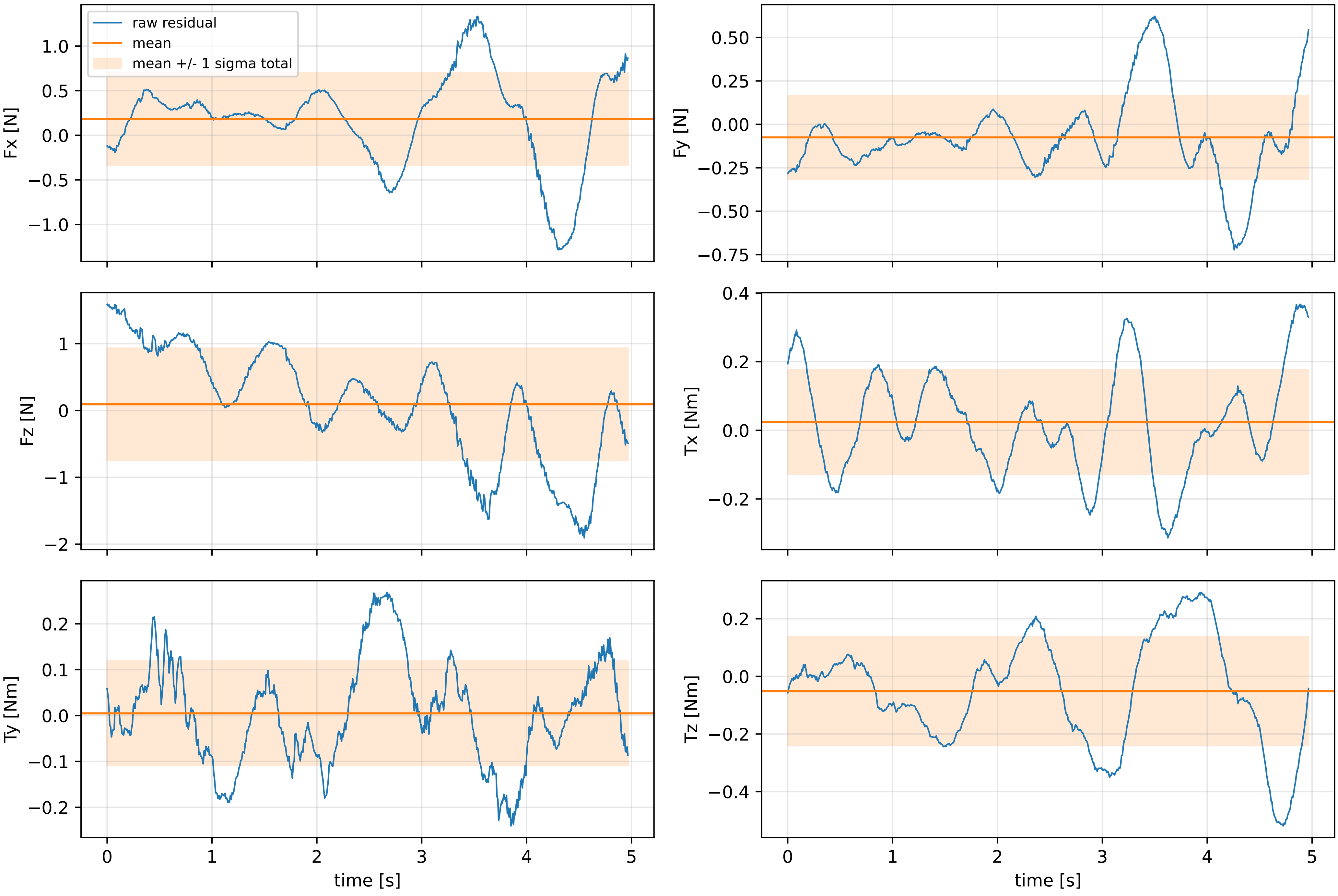
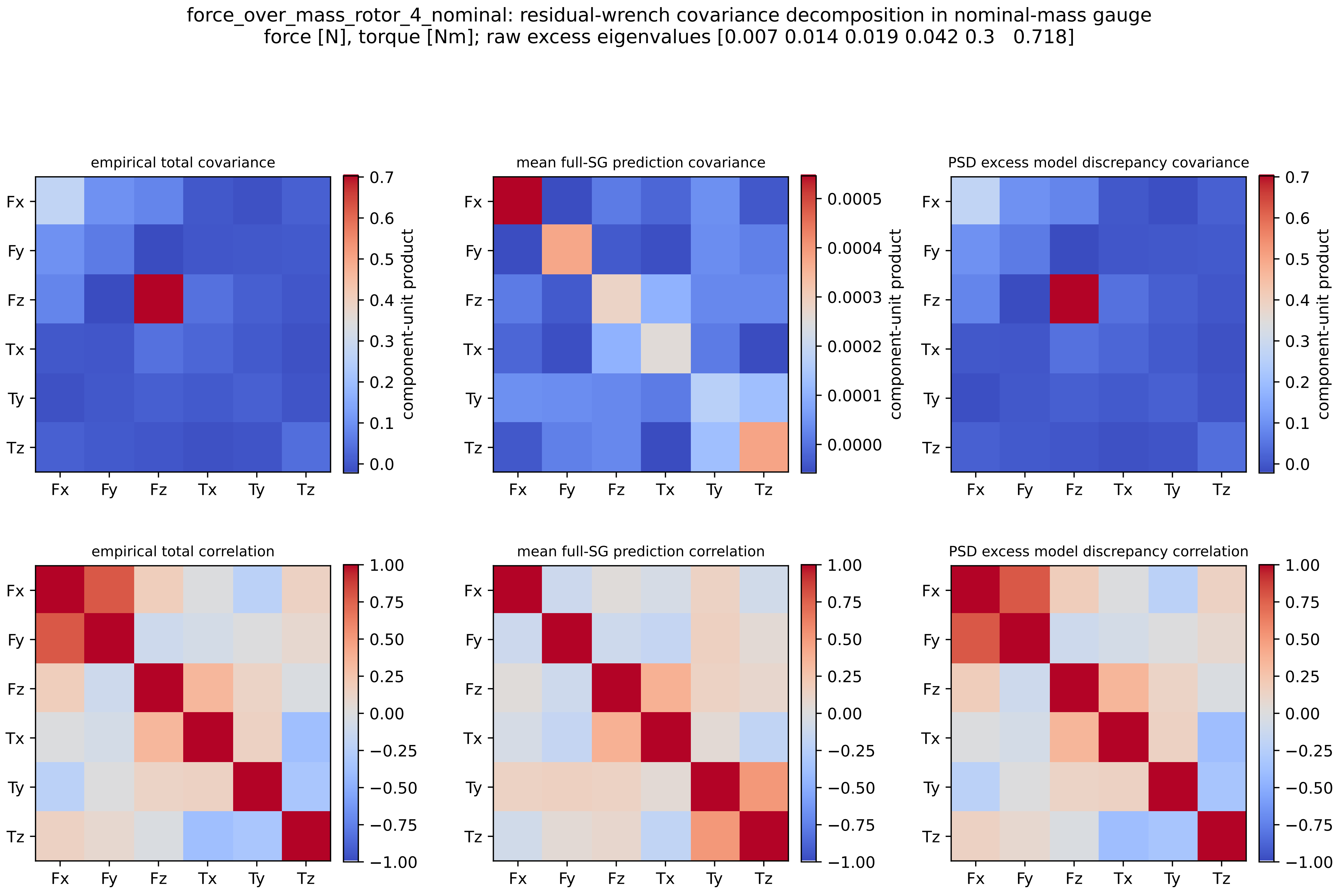
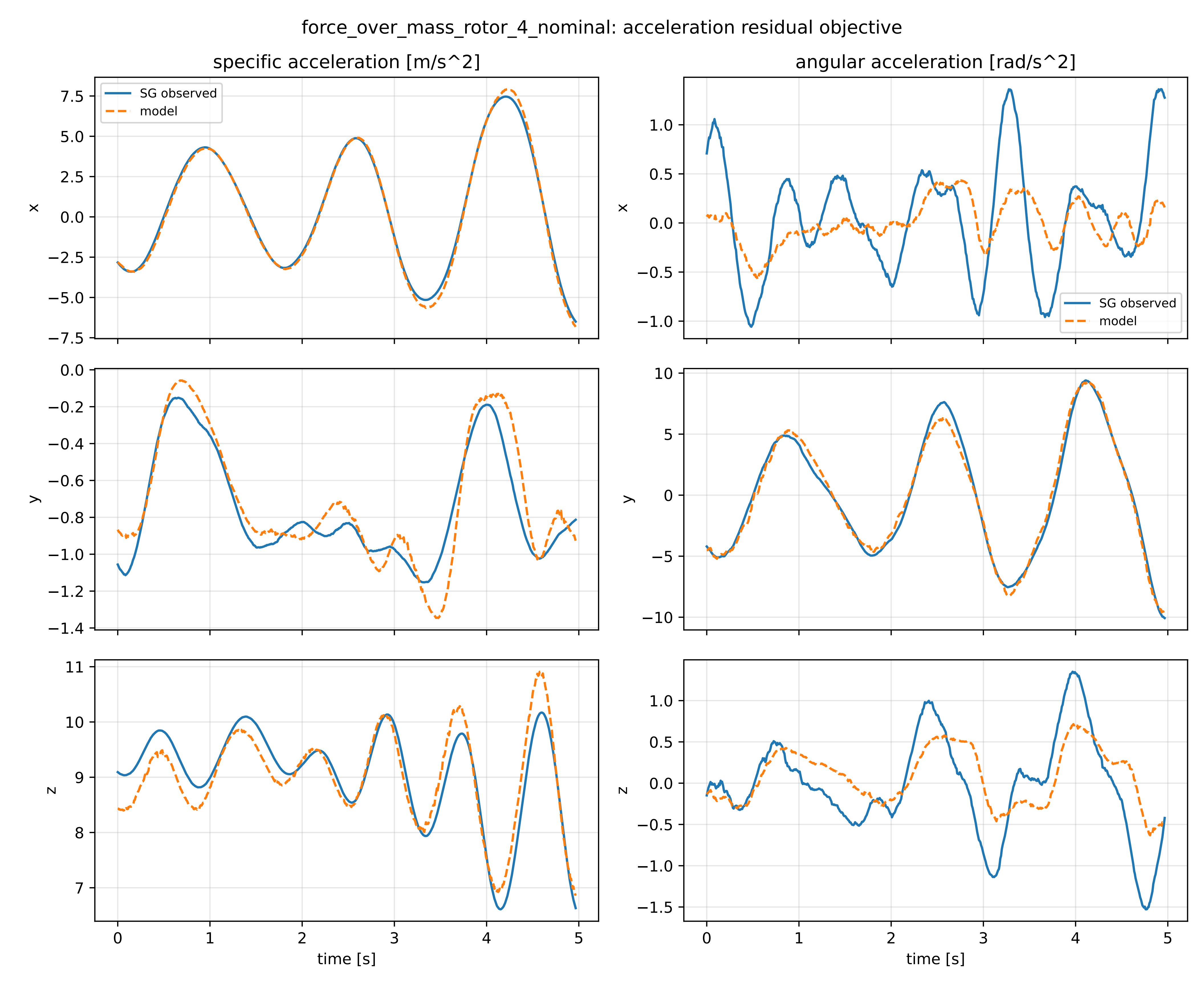

force_over_mass_rotor_4_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
2
1
0
−1
−2
10
5
0
−5
−10
0
1
2
3
4
5

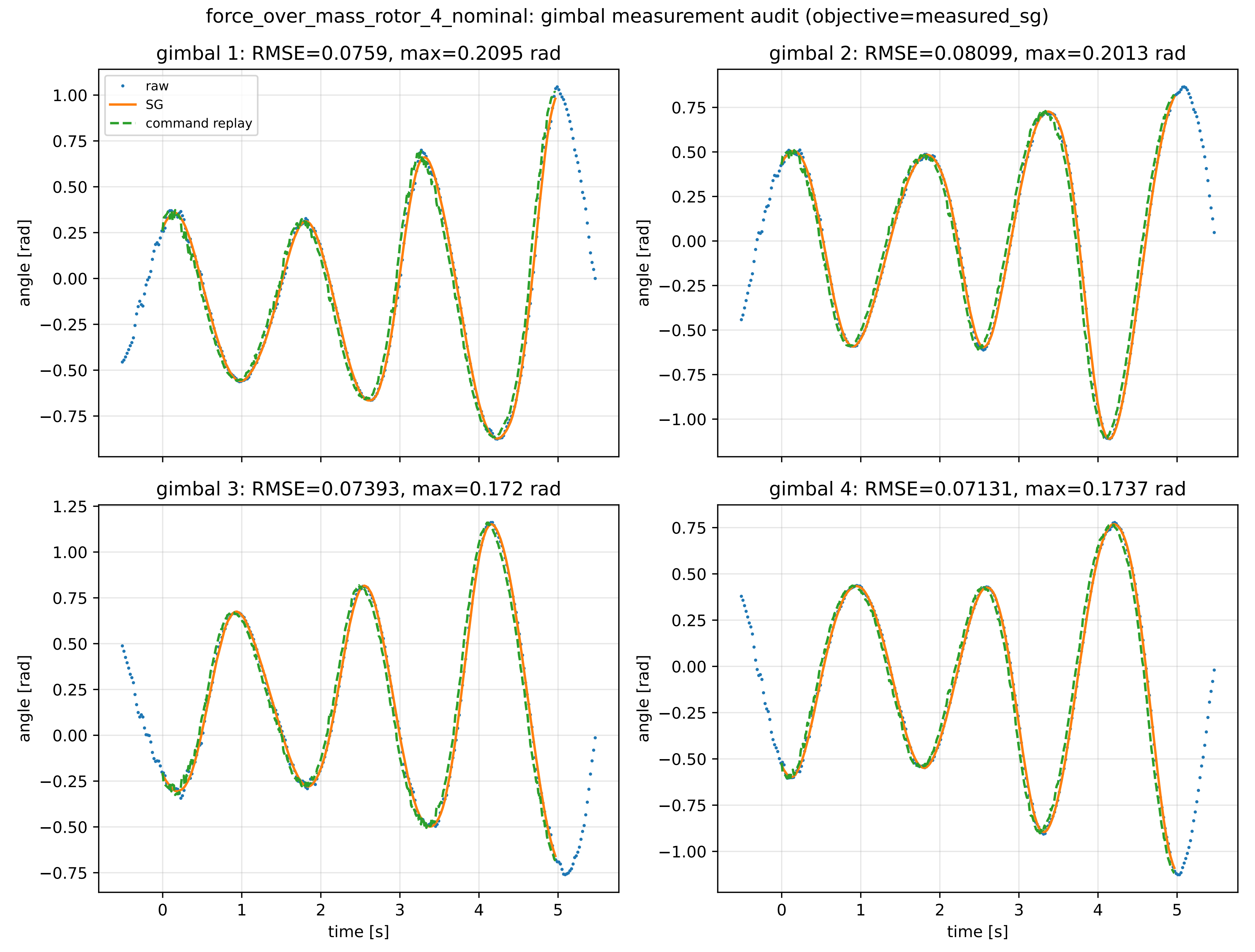
force_over_mass_rotor_4_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



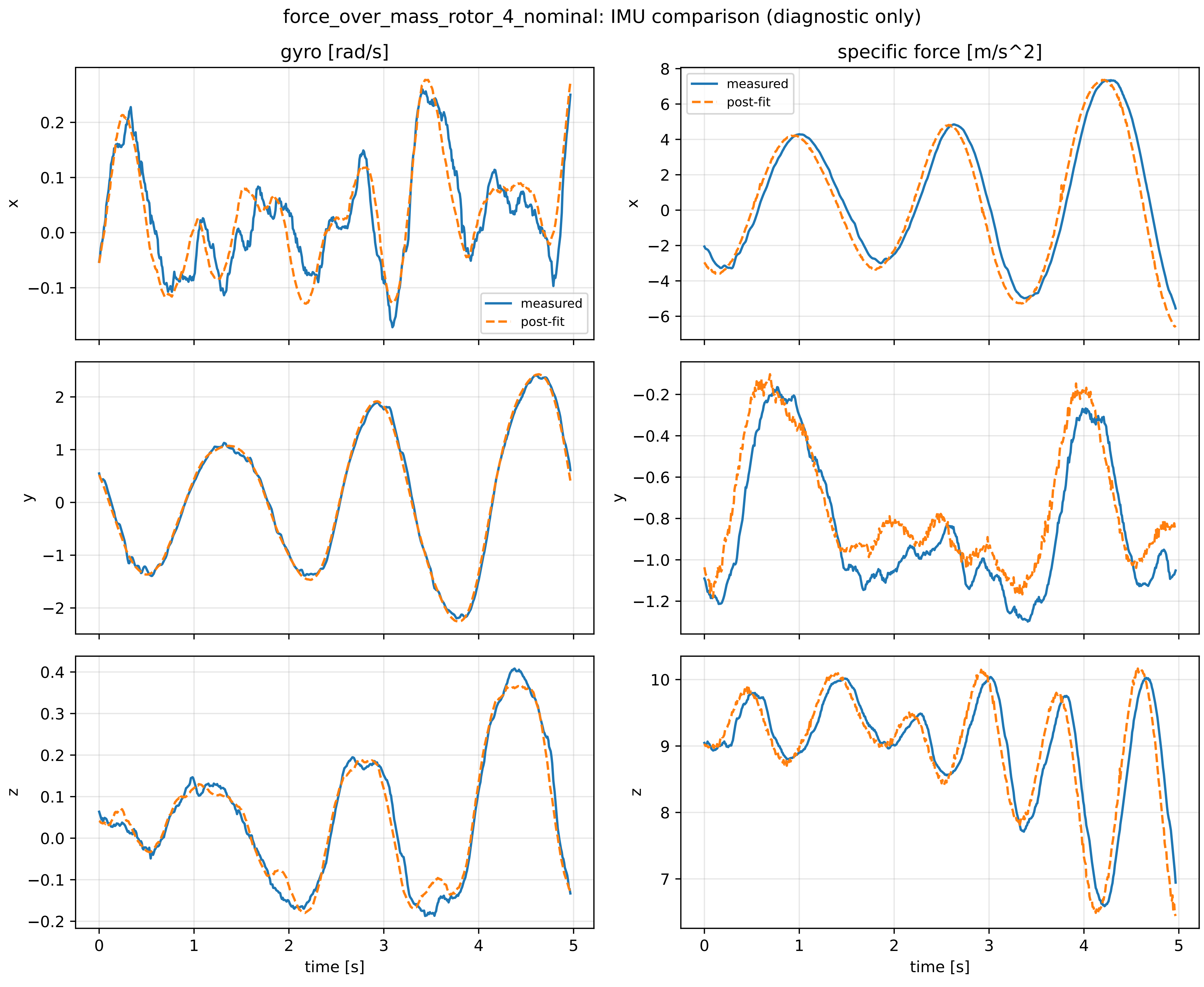
vector RMS about zero: force 1.038 N, torque 0.27287 Nm; component std about mean: force [0.5206 0.2405 0.8383], torque [0.1514 0.1135 0.1885]

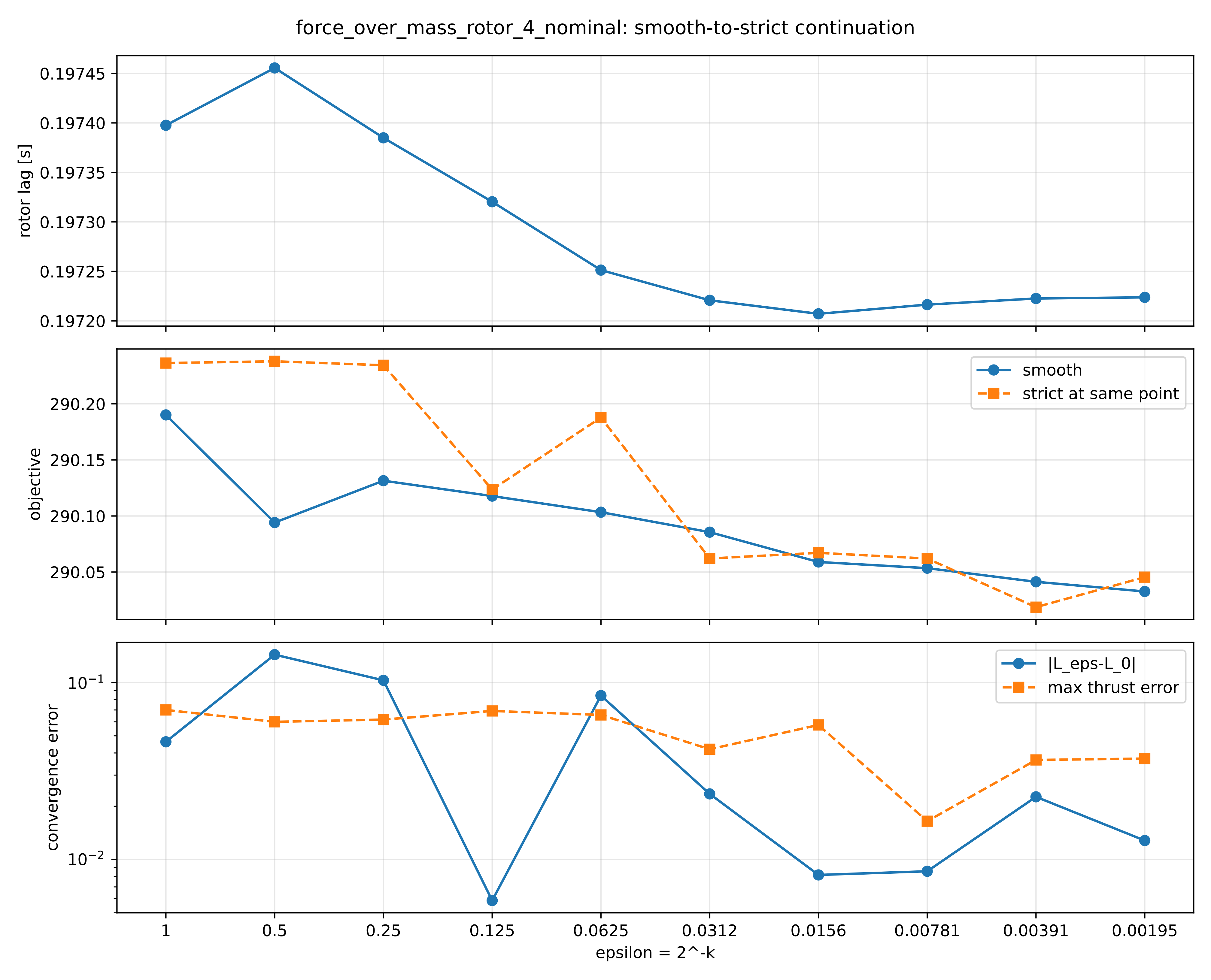

force_over_mass_rotor_4_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.007 0.014 0.019 0.042 0.3 0.718]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx
Fy
Fz
Tx
Ty
Tz


force_over_mass_rotor_4_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



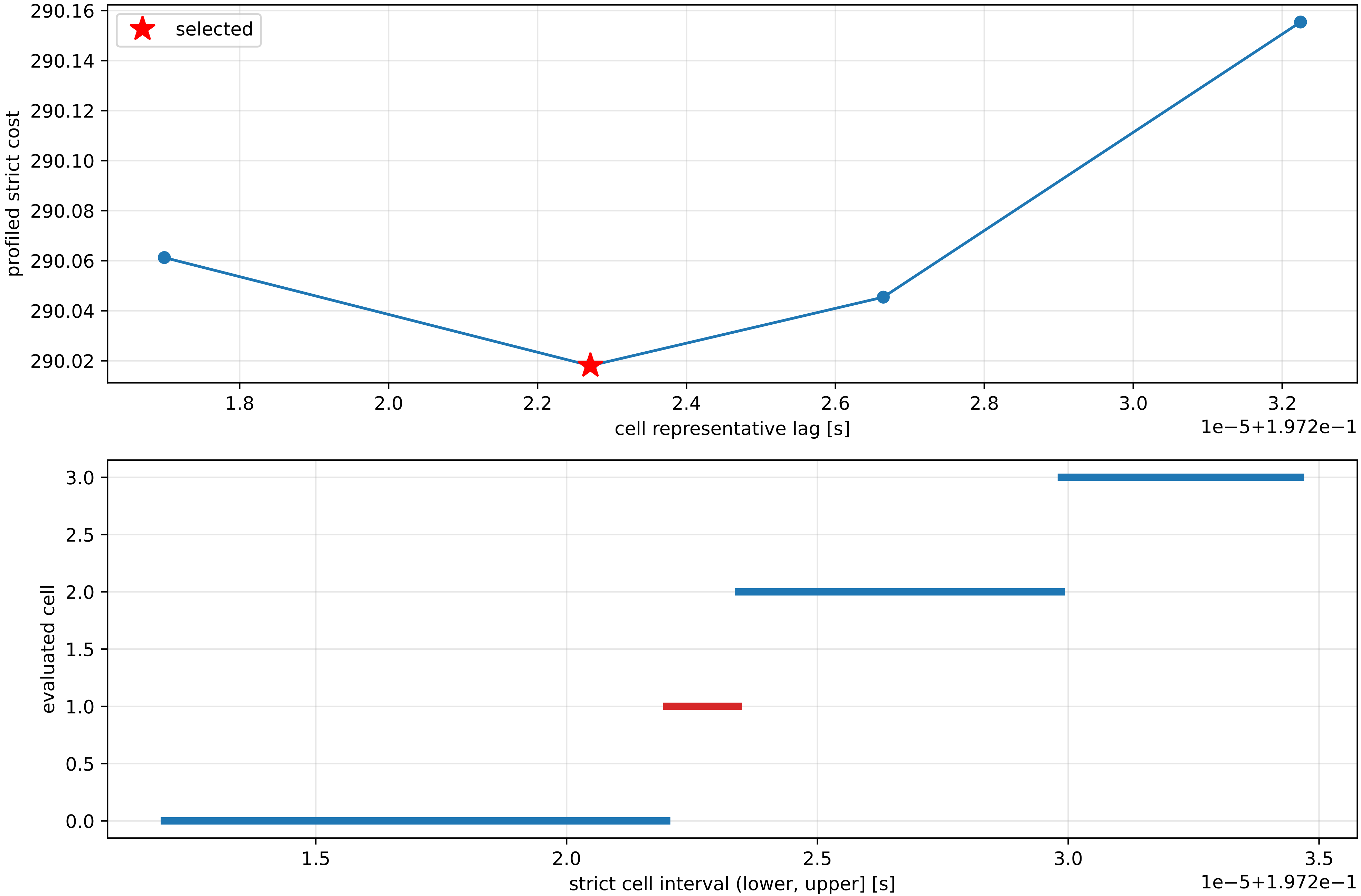
force_over_mass_rotor_4_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]


force_over_mass_rotor_4_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



force_over_mass_rotor_4_nominal: smooth-to-strict continuation
rotor lag [s]
0.19745
0.19740
0.19735
0.19730
0.19725
0.19720
objective
290.20
290.15
290.10
290.05
smooth
strict at same point
convergence error
10^-1
10^-2
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

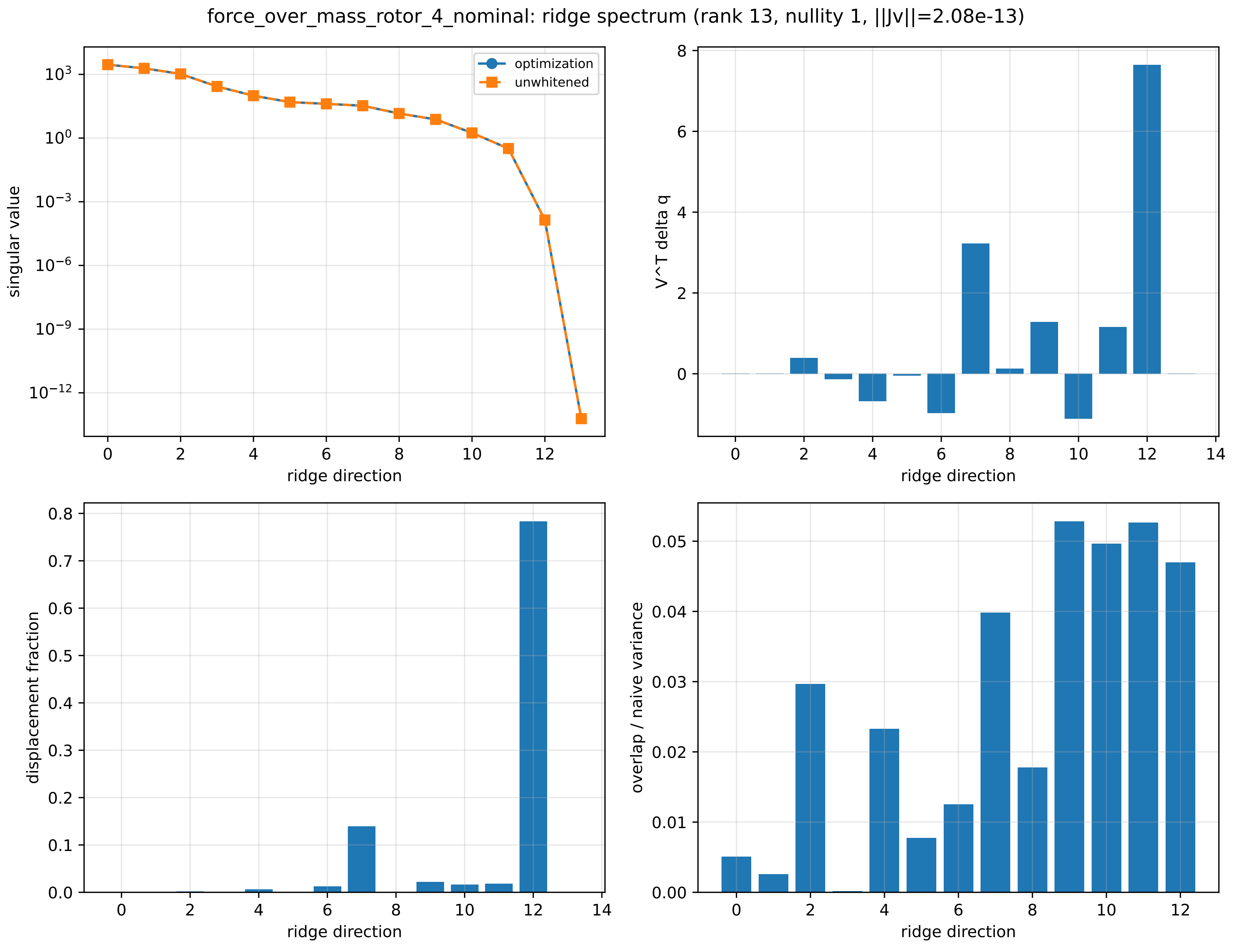
force_over_mass_rotor_4_nominal: exact strict-ZOH lag cells



selected (0.197221994, 0.197223425] s; final smooth 0.197223784 s; data boundary=False

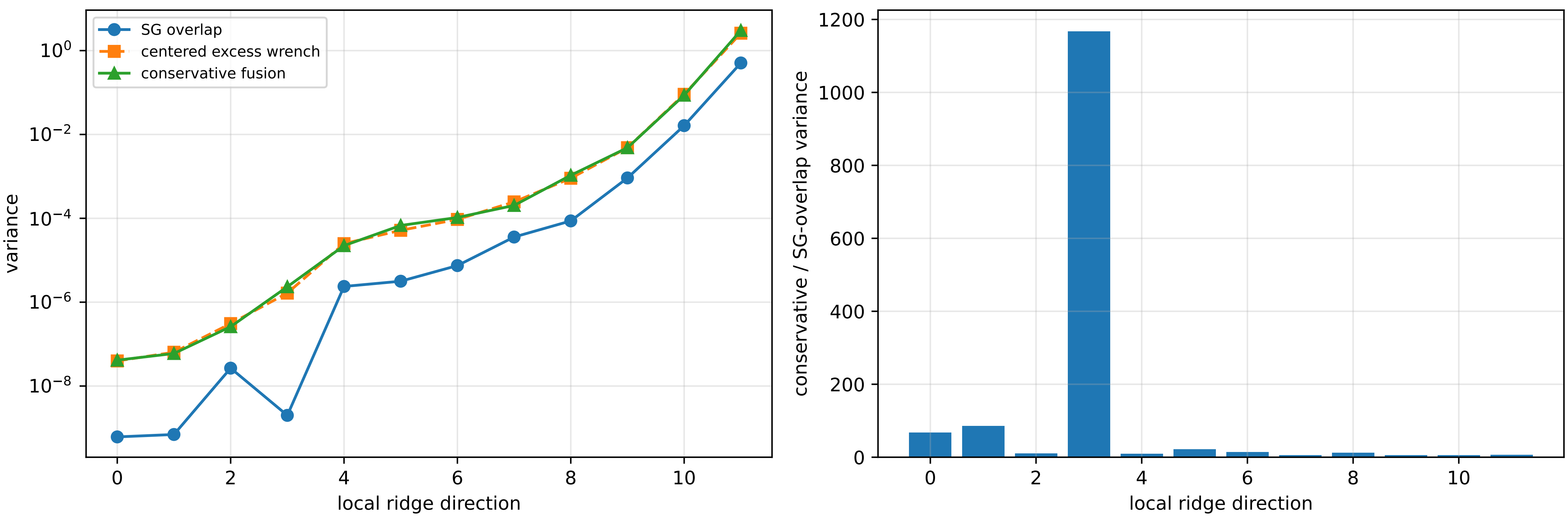
force_over_mass_rotor_4_nominal: parameter estimate

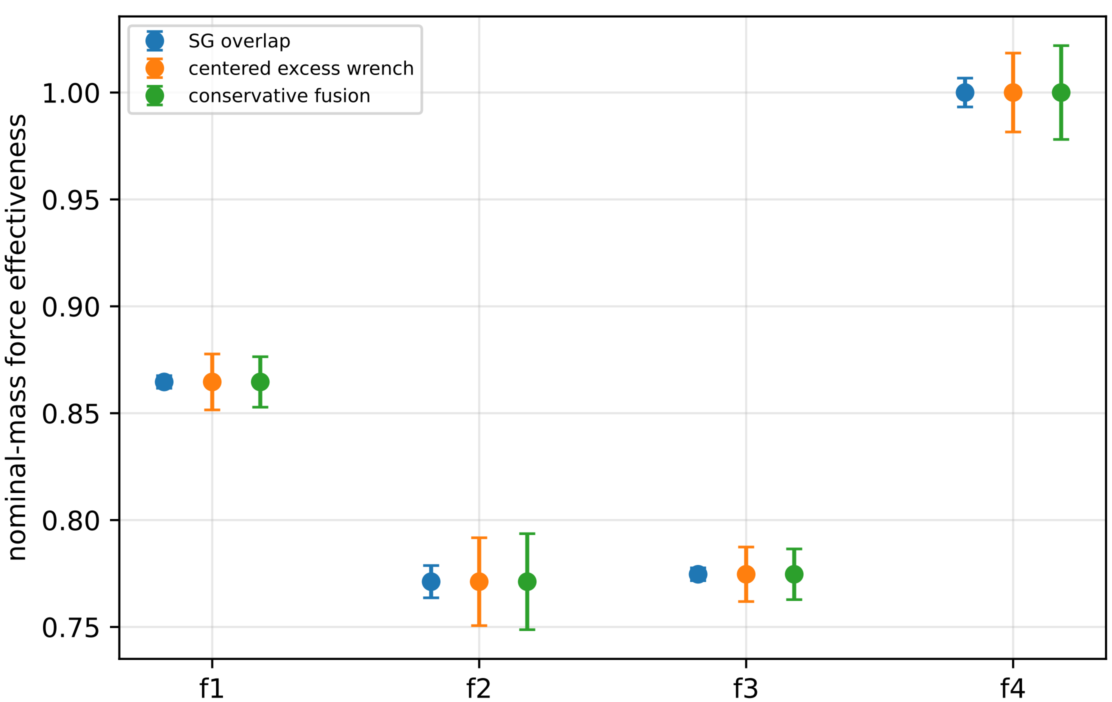
```
Case: force_over_mass_rotor_4_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.23442198
inertia [kg m^2]:
[[ 0.410852    0.00162315 -0.09222852]
 [ 0.00162315  0.26285481 -0.03473592]
 [-0.09222852 -0.03473592  0.5898924 ]]
CoG body [m]: [-0.02827708  0.01966923 -0.04904195]
force effectiveness: [1.18920344 1.06070309 1.06546186 1.37541463]

scale-free J/m [m^2]:
[[ 0.12702486  0.00050184 -0.02851468]
 [ 0.00050184  0.08126794 -0.01073945]
 [-0.02851468 -0.01073945  0.18237954]]
scale-free f/m: [0.36767108 0.32794209 0.32941337 0.42524279]

rotor lag cell: (0.197221994, 0.197223425] s
representative: 0.19722271 s
```



force_over_mass_rotor_4_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=2.08e-13)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction

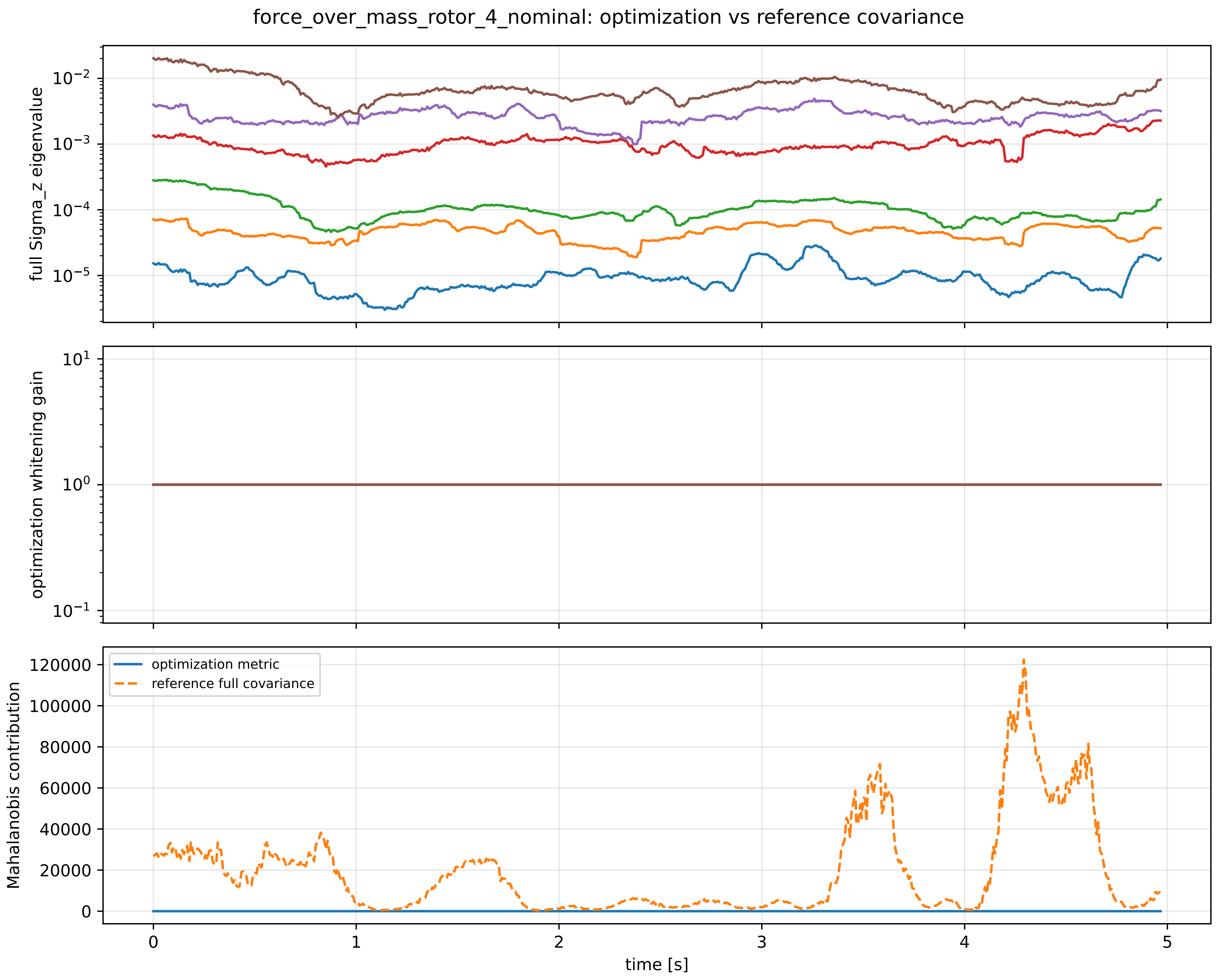
force_over_mass_rotor_4_nominal: Conservative fusion uncertainty

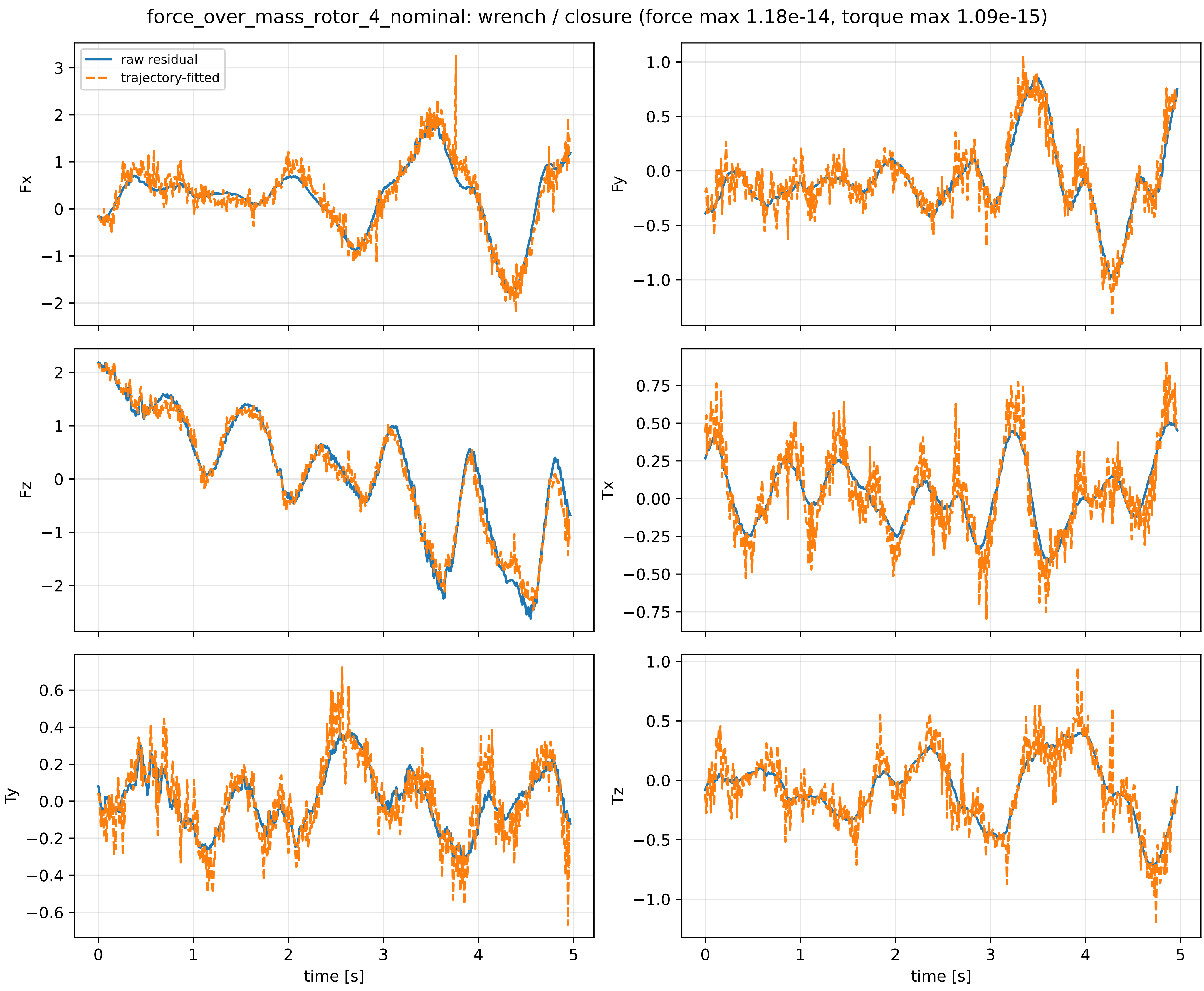





```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 46.05
wrench-to-acceleration recovery max error = 4.16e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=3.022, ratio=5.96
  log_mass=-0.0494; second_moment_diag_1=-0.366
  second_moment_diag_2=+0.161; second_moment_diag_3=+0.452
  second_moment_offdiag_12=+0.545; second_moment_offdiag_13=-0.479
  second_moment_offdiag_23=-0.312; cog_x=+9.86e-05
  cog_y=-0.0002; cog_z=+4.34e-05
  log_force_eff_1=-0.0492; log_force_eff_2=-0.0474
  log_force_eff_3=-0.0504; log_force_eff_4=-0.0503

index 10: variance=0.08635, ratio=5.316
  log_mass=+0.0358; second_moment_diag_1=-0.0638
  second_moment_diag_2=+0.315; second_moment_diag_3=-0.442
  second_moment_offdiag_12=+0.647; second_moment_offdiag_13=+0.513
  second_moment_offdiag_23=-0.0903; cog_x=+0.00273
  cog_y=-0.00804; cog_z=+0.00114
  log_force_eff_1=+0.0549; log_force_eff_2=+0.0858
  log_force_eff_3=+0.0139; log_force_eff_4=-8.74e-06

index 9: variance=0.004828, ratio=5.255
  log_mass=-0.157; second_moment_diag_1=+0.0202
  second_moment_diag_2=+0.826; second_moment_diag_3=-0.0125
  second_moment_offdiag_12=-0.229; second_moment_offdiag_13=-0.0997
  second_moment_offdiag_23=+0.268; cog_x=-0.0216
  cog_y=+0.00943; cog_z=-0.0062
  log_force_eff_1=-0.0925; log_force_eff_2=-0.319
  log_force_eff_3=-0.211; log_force_eff_4=-0.0544
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.


force_over_mass_rotor_4_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-2}
10^{-3}
10^{-4}
10^{-5}
optimization whitening gain
10^{1}
10^{0}
10^{-1}
Mahalanobis contribution
120000
100000
80000
60000
40000
20000
0
optimization metric
reference full covariance
0
1
2
3
4
5
time [s]



force_over_mass_rotor_4_nominal: wrench / closure (force max 1.18e-14, torque max 1.09e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

force_over_mass_rotor_4_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_force_over_mass_rotor_4_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass
SHA256: 678b0ccde06a50492269171f09bec596a62e0e49ae20754e771a2a2c2d4a9df2

data objective:  290.018058275
prior objective: 5.32197064926e-06
total objective: 290.018063597

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: force_over_mass_rotor_4_nominal (force_effectiveness_over_mass)
  components: rotor_4
  target: [0.42524282]
  std:    [1.e-05]
  final:  [0.42524279]
  error:  [-3.26250537e-08]
  standardized residual: [-0.00326251]
  factor objective: 5.32197064926e-06
```